EXPERIMENT 3

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Semester: 5 **Subject Code: 24CSH-337**

Subject Name: Computer Network

Aim - To study the functionality of a Hub, Switch, and Router by designing and simulating different network topologies in Cisco Packet Tracer, and to examine the communication process between the connected devices.

Objective

- To study the operating principles of hubs, switches, and routers, and understand their significance in computer networks.
- To design and simulate network topologies in Cisco Packet Tracer by interconnecting end devices using hubs, switches, and routers.
- To assign and configure IP addresses and routing settings to establish communication within a single network as well as across multiple networks.
- To evaluate and compare the communication process through hubs, switches, and routers by monitoring packet transmission and network performance in Cisco Packet Tracer's simulation mode.

Theory:

- **<u>HUB</u>**:
  - **A hub operates at the Physical Layer (Layer 1) of the OSI model.**
    It is responsible only for transmitting electrical or digital signals between connected devices. Since it works at the physical layer, it does not examine or process the data being transmitted.
  - **A hub is primarily used to establish a Local Area Network (LAN).**
    It connects multiple computers or network devices within a limited geographical area, enabling them to communicate with one another over the same network.
  - **A hub contains multiple ports for connecting different network devices.**
    Each port serves as an interface for devices such as computers, printers, or servers. This allows several devices to share the same communication medium.
  - **A hub is commonly used to create a star topology.**
    In this topology, all devices are connected to a central hub, making installation and expansion easier. However, if the hub fails, communication between all connected devices is interrupted.
  - **When a data packet reaches one port, the hub broadcasts it to all other connected ports.**
    It does not identify the intended destination or filter traffic, so every connected device receives the same packet. This broadcasting behavior can increase network congestion and reduce overall efficiency compared to modern networking devices like switches.

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- **SWITCH :**
  - **A switch is a networking device that connects multiple devices to form a Local Area Network (LAN).**
    It enables efficient communication between computers, printers, servers, and other network devices by forwarding data only to the intended destination.
  - **Unlike a hub, a switch has memory to store network information.**
    It learns and maintains the MAC addresses of connected devices, allowing it to make intelligent forwarding decisions instead of broadcasting data to every port.
  - **A switch maintains a MAC Address Table (CAM Table).**
    This table maps the MAC address of each connected device to a specific switch port. As a result, data frames are sent only through the appropriate port, improving network performance and reducing unnecessary traffic.
  - **A switch operates at the Data Link Layer (Layer 2) of the OSI model.**
    It processes Ethernet frames and uses MAC addresses to determine the correct destination device, making communication faster and more secure than a hub.
  - **A switch contains multiple interfaces or ports, each identified by a unique port number.**
    Every connected device is associated with a specific port in the MAC Address Table, making it easier to identify which device is connected to which interface and to manage the network efficiently.

- **Router**:
  - **A router is a networking device that forwards data packets between different computer networks.**
    It is mainly used to connect two or more separate LANs or to link a LAN with a Wide Area Network (WAN), allowing communication between different networks.
  - **A router is connected to at least two different networks.**
    These connections may include two LANs, two WANs, or a LAN and an Internet Service Provider (ISP) network. It acts as a gateway for transferring data from one network to another.
  - **A router operates at the Network Layer (Layer 3) of the OSI model.**
    It uses IP addresses to determine the best path for forwarding data packets to their destination, ensuring efficient and reliable communication across networks.
  - **A router maintains a Routing Table in its memory to store network routes.**
    The routing table contains information about available network paths and helps the router select the most appropriate route for transmitting packets. This improves network efficiency and ensures accurate packet delivery.

Practical/Experiment Steps

HUB:

STEPS:

1. **Open Cisco Packet Tracer and select a Hub from the Network Devices section.**
Drag and drop the hub into the workspace. Click on the hub to view the available FastEthernet ports that will be used for co

2 Add three PCs to the workspace.
From the **End Devices** section, drag and place three PCs near the hub to create the required network topology

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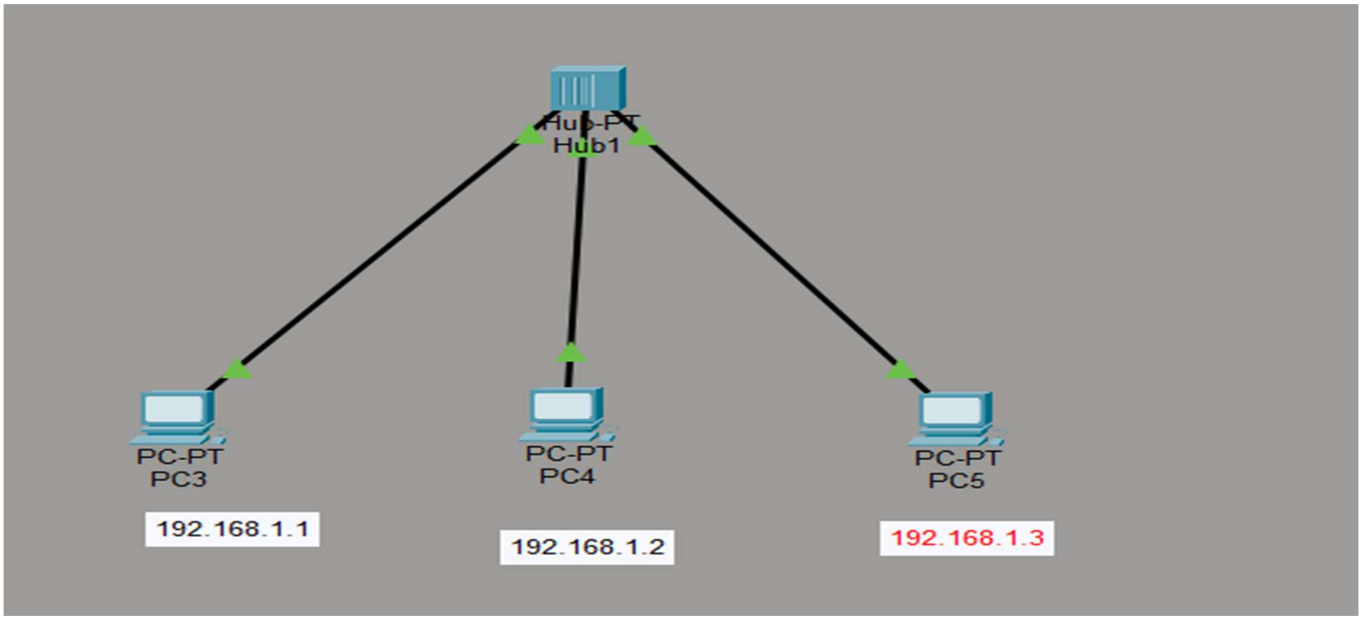
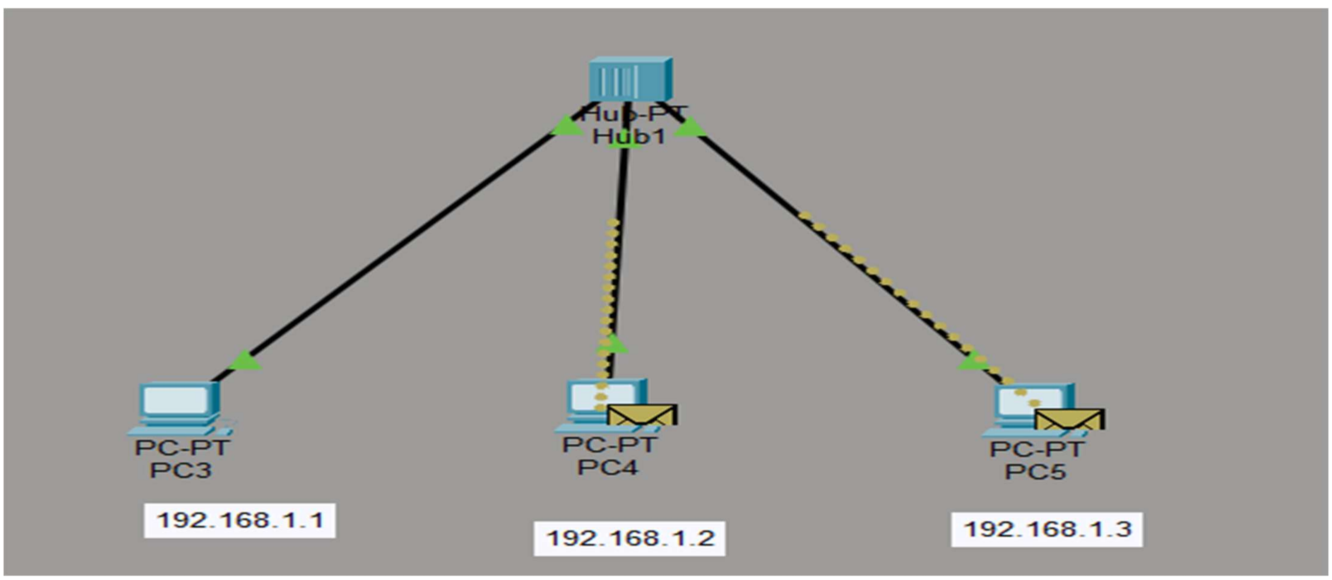
3.Connect each PC to the hub using Copper Straight-Through cables.
Select the **Copper Straight-Through** cable, connect the **FastEthernet0** port of each PC to any available FastEthernet port on the hub. After all connections are made, verify that the link indicators turn green, indicating successful connectivity.



[pic 1.1]

4. Switch to Simulation Mode by clicking the *Simulation* option located at the bottom-right corner of Cisco Packet Tracer.
Create a **Simple PDU (data packet)** and set **192.168.1.2** as the source IP address and **192.168.1.3** as the destination IP address. Since the devices are connected through a **hub**, the packet is broadcast to all connected ports, including the device with IP address **192.168.1.2**, even though it is not the intended recipient, as illustrated in **Figure 1.2**.



[pic 1.2]

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Advantage:

1. Cheaper than Switches.

2. Works good for smaller networks.

Disadvantages:

1. Issues with broadcasting.

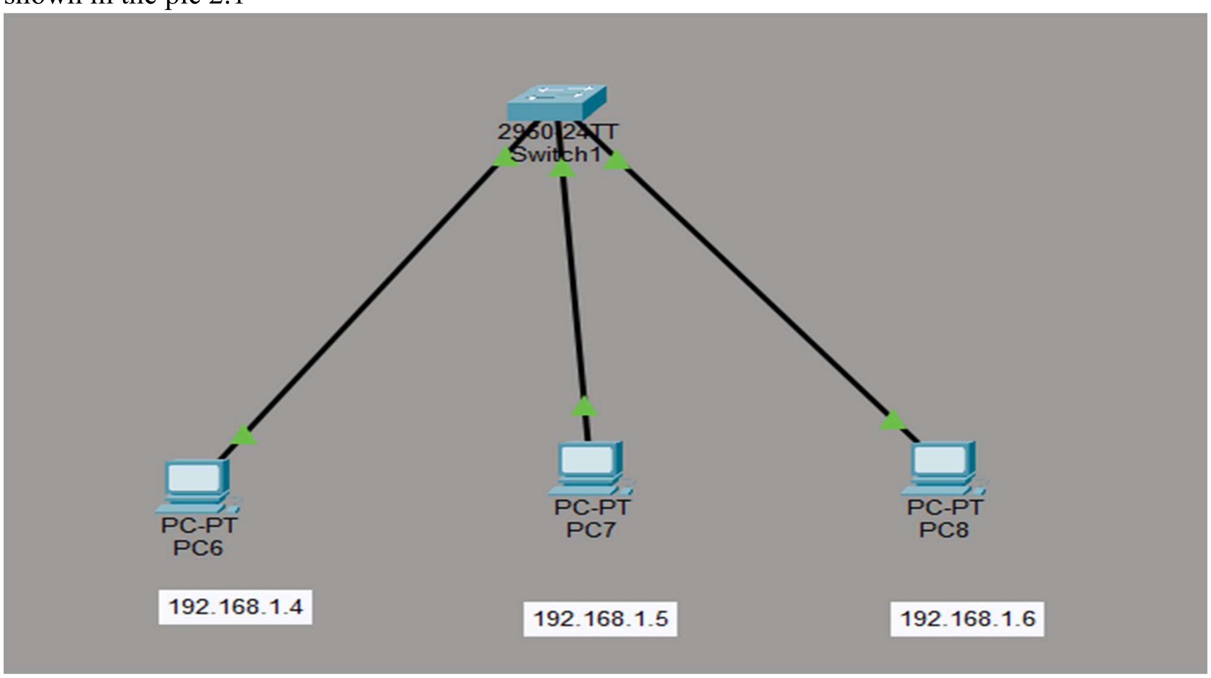
2. No Memory.

3. Normally rins in half-duplex mode.

<u>Switches</u>:

Steps:

1. Switch section > choose one switch > click on switch and check for ports
2. Take 3 PC's and take straight through cable -> Connect all PC's with Switch with FastEthernet port as shown in the pic 2.1



[pic 2.1]

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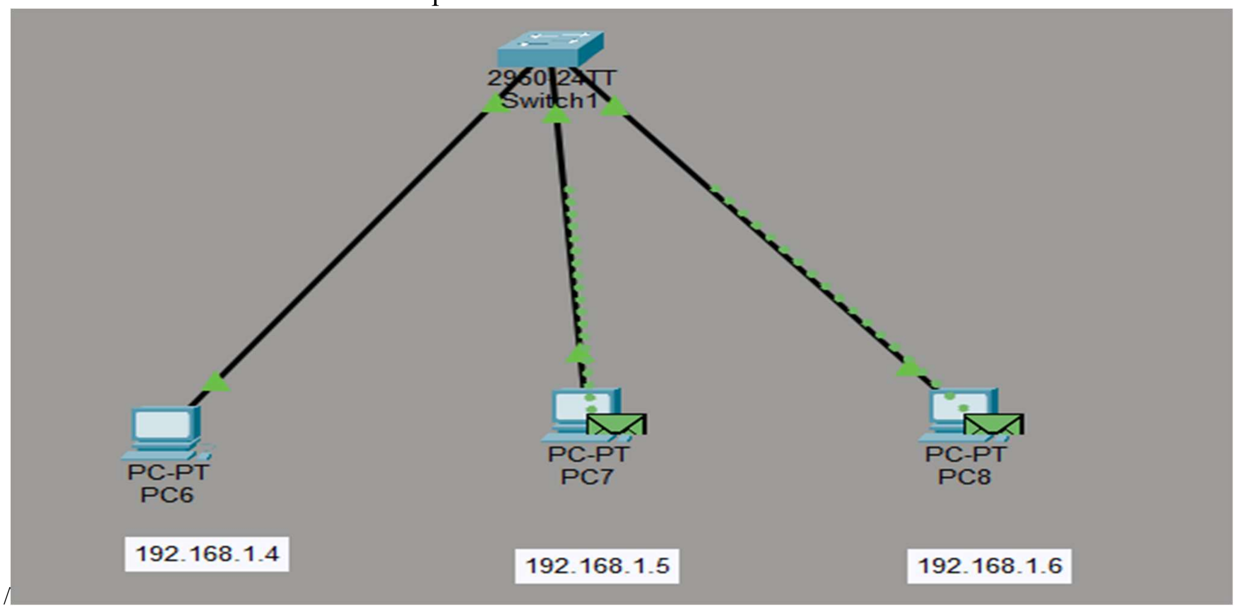
3. On bottom right corner click on simulation -> take one data packet -> take 192.168.1.1 as Source_ip & 192.168.1.3 as Destination_ip and unlike hub it will not be broadcasted to all the other PCs and only be send to destination PC like shown in the pic 2.2



[pic 2.1]

Router:

Steps

1. Create two LANs using a Switch:

- LAN 1: IP 10.0.0.0 - Subnet Mask: 255.0.0.0

- Take 1 Switch & 3 PC's

- Configure the IPs of each PC as 10.0.0.1 for 1st PC, 10.0.0.2 for 2nd PC, 10.0.0.3 for 3rd PC

- Check this LAN 1 by sending a a PDU from 10.0.0.1 to 10.0.0.3.

2. Create 2nd LANs using a Switch:

- LAN 2: IP 20.20.20.0 - Subnet Mask: 255.255.255.0

- Take 1 Switch & 3 PC's

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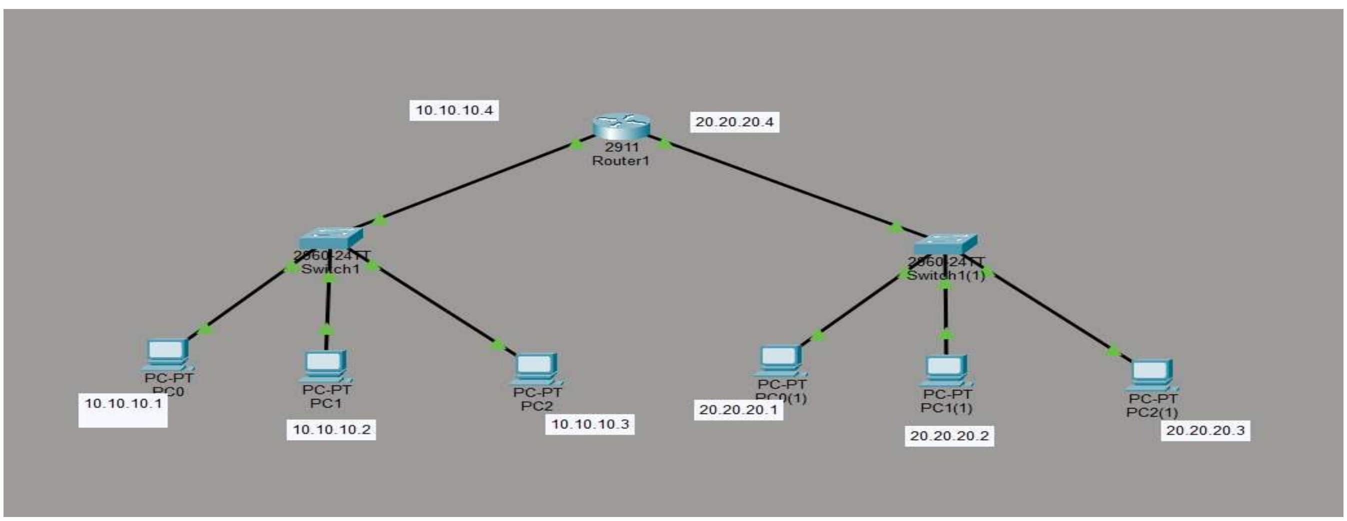
- Configure the IPs of each PC as 192.168.1.1 for 1st PC, 192.168.2 for 2nd PC, 192.168.3 for 3rd PC
- Check this LAN 2 by sending a a PDU from 20.20.20.1 to 20.20.20.3.

3. Take one router (any router like Router 2911)
- Now this router has 3 interfaces: GigabitEthernet 0/0, GigabitEthernet 0/1, and GigabitEthernet 0/2 which means, we can connect three different LANs using these interfaces.
- Connect switch from LAN 1 with GigabitEthernet 0/1 with Router's GigabitEthernet 0/0.
- Connect switch from LAN 2 with GigabitEthernet 0/1 with Router's GigabitEthernet 0/1.

4. Now to send data from PC0 to PC5 we need to configure the default gateway as well for both LAN's.
-For LAN1: Gateway Address: 10.0.0.4
-For LAN2: Gateway Address: 20.20.20.4
- Now go to every PC of each LAN and configure the Gateway Address to each PC of each LAN.
[Click PC0 -> Go to Desktop -> IP Configuration -> Enter the Gateway Address in Default Gateway.



[pic 3.1]

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Learning Outcome

- **Understand and distinguish the roles of a Hub, Switch, and Router in a computer network.**
  The learner will be able to explain the working principles, features, and differences between these networking devices and identify their appropriate applications.
- **Design and simulate network topologies using Cisco Packet Tracer.**
  The learner will be able to create basic LAN topologies by connecting end devices through hubs, switches, and routers, and verify successful network connectivity.
- **Configure IP addressing and routing to enable network communication.**
  The learner will be able to assign IP addresses, configure routing settings, and establish communication both within the same network and between different networks.
- **Evaluate network performance through packet analysis in simulation mode.**